Alessandro Donina

MSc Aerospace Engineering | Aerodynamics and Aeroacoustics
Turin, Italy | alessandrodonina@gmail.com | +39 351 351 2286 | linkedin.com/in/alessandro-donina

PROFILE

MSc Aerospace Engineering student at Politecnico di Torino, Italy.
Passionate about aeronautics, I developed a strong interest in aerodynamics during my undergraduate studies, which led me to pursue a Master's degree specializing in this field. From March 2025 to October 2026, I have been part of the ZEFIRO Student Team, applying and expanding the knowledge gained through my studies.
Experienced in aerodynamic design, optimization and analysis of eVTOLs for Urban Air Mobility solutions.
Hands-on experience with Ansys Fluent, Siemens STAR-CCM+, OpenVSP, MATLAB, Python, xfoil, XFLR5 and DFDC.

EDUCATION

MSc in Aerospace Engineering (LM-20) — Politecnico di Torino March 2025 – Present
GPA 27.79 /30 (Italian scale 18–30, 30 = highest honours)
Relevant coursework: Unsteady Aerodynamics; Aeroelasticity; Computational Fluid Dynamics (CFD); Turbulent Flows; CFD for Propulsion Systems; Experimental Aerodynamics; Aeroacoustics.

BSc in Aerospace Engineering — Politecnico di Torino September 2021 – March 2025
Thesis: "Numerical Solution of Convective Heat Transfer over a Finite Flat Plate" (Thermo-fluid Dynamics).

EXPERIENCE

Chief Engineer — Team ZEFIRO, Politecnico di Torino March 2026 – Present
Leading the technical division of the student team (70+ people), overseeing both the design and build of Urban Air Mobility solutions and contributing to the administrative efforts of the entire team.

- **Projects:**
  - VESPA Mk1 - Agricultural eVTOL: ongoing project, involved in final design optimization (e.g., winglet design, battery selection, parachute fairing), PID tuning and prototype assembly;
  - Airspeeder – Electric Racing eVTOL: contributed to drafting administrative and participation agreements, developed the initial workflow for the aerodynamic and structural optimization of the Mk3c prototype.

Aerodynamic Engineer — Team ZEFIRO, Politecnico di Torino March 2025 – March 2026
Aerodynamic design, optimization and analysis of eVTOLs for Urban Air Mobility solutions.

- **Projects:**
  - RED Ferraris - Hydrogen-Powered VTOL for the VFS Competition 2025 ("Best New Entrant" Prize Winner): developed a workflow to generate an optimized Ducted Propeller validated through CFD, achieving 700 N of thrust with a 0.72 meter diameter ducted propeller;
  - VESPA Mk1 - Agricultural eVTOL: propulsion system selection, CFD validation of forced boundary layer transitions using LES analysis, eVTOL simulation testing on X-Plane 11 (dynamics and stability of the eVTOL).

HONOURS & CERTIFICATIONS

1st place – Flight Mechanics Contest, 2nd edition (team) — DIMEAS, Politecnico di Torino Jan 2026
Aircraft design competition with OpenVSP and MATLAB in the new in-class laboratory format.
Won the competition as the most efficient aircraft design in terms of flight stability.

TECHNICAL SKILLS

- **CAD & design:** SolidWorks, Siemens NX, OpenVSP (aircraft conceptual design);
- **Low-fidelity Analysis:** xfoil, XFLR5, DFDC;
- **CFD Software:** Ansys Fluent, Siemens STAR-CCM+;
- **Programming:** MATLAB, Python, Fortran (basic).

LANGUAGES

Italian: Native | **English:** IELTS 7.5 (C1)